



Saint Columban College
College of Computing Studies
Pagadian City



Tab, Tab, Attack

Tab, Tab, Attack 



Easy

General Skills

picoCTF 2021

AUTHOR: SYREAL

Description

Using tabcomplete in the Terminal will add years to your life, esp. when dealing with long rambling directory structures and filenames.

[Addadshashanammu.zip](#)

Hints 

1

112,172 users solved



85%

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picoCTF{FLAG}

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Author: The Analyst: Hyposelenia

Challenge: Tab, Tab, Attack

Category: General Skills

Date: 01/11/26



I. Objective

The objective of this task was to identify the hidden flag by unzipping a provided ZIP file and navigating through a deeply nested directory structure. Rather than manually typing long file and folder names or even opening one-by-one, the challenge emphasized the effective use of shell tab auto-completion to quickly and efficiently reach the file containing the flag.

II. Background

In Linux systems, files and folders are organized in hierarchical directory structures. Navigating these directories using the command line can become inefficient when paths and filenames are long or complex. To address this, Unix-like shells provide a feature called tab auto-completion, which allows users to automatically complete filenames or directory names by pressing the Tab key. This challenge was designed to introduce beginners to this essential productivity feature, demonstrating how proper command-line navigation can significantly reduce effort and errors.

III. Tool Used

Oracle VirtualBox (Kali Linux) – used as the Linux environment required to run the challenge

Linux Terminal – used to execute file and navigation commands

unzip – used to extract the contents of the ZIP archive

ls – used to list files and directories

cat – used to display the contents of the file containing the flag

Tab auto-completion – used to quickly complete long directory and file names

IV. Methodology

1. The ZIP file provided in the challenge was downloaded.
2. Kali Linux was launched using Oracle VirtualBox.
3. The ZIP file was located in the system using the **ls** command.
4. The archive was extracted using the command **unzip <filename>.zip**.
5. After extraction, a directory containing multiple nested folders with long names was created.



6. Instead of manually typing each directory name, the Tab key was used to automatically complete folder and file names.
7. At the deepest level of the directory, two files were present.
8. The correct file was identified and opened using the **cat** command with tab auto-completion, revealing the flag.

V. Result

picoCTF{l3v3l_up!_t4k3_4_r35t!_fc588427}

```
File Actions Edit View Help
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ ls
Addadshashanammu.zip  flag  warm

(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ unzip Addadshashanammu.zip
Archive: Addadshashanammu.zip
  creating: Addadshashanammu/
  creating: Addadshashanammu/Almurbalarammi/
  creating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/
  creating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/
  creating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/
  creating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/Onnissiralis/
  creating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/Onnissiralis/Ularradallaku/
  extracting: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/Onnissiralis/Ularradallaku/fang-of-haynekhtnamet.c
  inflating: Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/Onnissiralis/Ularradallaku/fang-of-haynekhtnamet

(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$

(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ cat Addadshashanammu/Almurbalarammi/Ashalmimilkala/Assurnabitashpi/Maelkashishi/Onnissiralis/Ularradallaku/fang-of-haynekhtnamet.c
#include <stdio.h>

int main(){
printf("ZAP!* picoCTF{l3v3l_up!_t4k3_4_r35t!_fc588427}\n");
}

(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$
```

VI. Explanation

This challenge focused on efficient command-line navigation rather than file analysis or decoding. The ZIP file contained a deeply nested directory structure designed to discourage manual exploration. By relying on tab auto-completion, the shell automatically filled in directory and file names after a few keystrokes, allowing rapid traversal of the file system. This approach minimized typing errors and significantly reduced the number of keystrokes required, aligning with the challenge hint that the solution could be achieved with only a small number of button presses.



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VII. Conclusion

The challenge highlighted the importance of understanding basic Linux file system operations such as unzipping archives, navigating directory paths, and viewing file contents. Most importantly, it demonstrated the practical value of tab auto-completion as a powerful tool for improving efficiency in the command line. Mastery of these foundational skills is essential for effective work in cybersecurity, system administration, and Linux-based environments.

— **The Analyst: Hyposelenia**